

Precalculus Review for Calculus

ANSWER KEY



Functions and composition

- 1 Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$. Find:
- a** $(f \circ g)(x)$ $f(g(x)) = \sqrt{x^2 + 2}$.
- b** $(g \circ f)(x)$ $g(f(x)) = (x-1) + 3 = x + 2$, for $x \geq 1$.
- c** the domain of $f \circ g$ $x^2 + 2 \geq 1$ always holds, so the domain is all real numbers.
- 2 Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = x^2 - 3x$.
- $$\frac{(x+h)^2 - 3(x+h) - (x^2 - 3x)}{h} = \frac{2xh + h^2 - 3h}{h} = 2x + h - 3.$$

Lines, exponentials, and trig — the toolkit calculus builds on

- 3 Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.
- Slope $= \frac{-2-4}{3-(-1)} = -\frac{3}{2}$; equation $y - 4 = -\frac{3}{2}(x + 1)$, i.e. $y = -\frac{3}{2}x + \frac{5}{2}$.
- 4 Solve $2e^{3x} = 14$ exactly and to three decimal places.
- $e^{3x} = 7$, so $x = \frac{\ln 7}{3} \approx 0.649$.
- 5 State the exact value of each expression, needed frequently when differentiating trig functions:
- a** $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ **b** $\cos \frac{\pi}{3} = \frac{1}{2}$ **c** $\tan \frac{\pi}{6} = \frac{\sqrt{3}}{3}$

Factoring and rational expressions

- 6 Factor each expression fully:
- a** $x^2 - 5x + 6 = (x-2)(x-3)$ **b** $4x^2 - 9 = (2x-3)(2x+3)$ **c** $x^3 - 8 = (x-2)(x^2 + 2x + 4)$
- 7 Simplify $\frac{x^2-4}{x^2-x-6}$, stating the value(s) of x excluded from the domain.
- $\frac{(x-2)(x+2)}{(x-3)(x+2)} = \frac{x-2}{x-3}$, excluding $x = -2$ (removable) and $x = 3$ (undefined).

Inverse functions and logarithms

- 8 Find the inverse of $f(x) = 3x - 5$, and verify by computing $f(f^{-1}(x))$.
- Swap and solve: $x = 3y - 5 \Rightarrow y = \frac{x+5}{3}$, so $f^{-1}(x) = \frac{x+5}{3}$. Check:
- $f(f^{-1}(x)) = 3 \cdot \frac{x+5}{3} - 5 = x + 5 - 5 = x$. ✓

- 9 Solve $\ln(x - 1) = 2$ exactly and to three decimal places.

$$x - 1 = e^2, \text{ so } x = e^2 + 1 \approx 8.389.$$

Piecewise and absolute value review

- 10 Sketch-reasoning: rewrite $f(x) = |2x - 6|$ as a piecewise function without absolute value bars.
 $f(x) = 2x - 6$ for $x \geq 3$, and $f(x) = 6 - 2x$ for $x < 3$ (since $2x - 6 \geq 0$ exactly when $x \geq 3$).
- 11 Solve the inequality $|3x + 1| \leq 7$, writing your answer in interval notation.
 $-7 \leq 3x + 1 \leq 7$: $-8 \leq 3x \leq 6$: $-\frac{8}{3} \leq x \leq 2$. Interval: $[-\frac{8}{3}, 2]$.
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Systems of equations

- 12 Solve the system $2x + y = 7$ and $x - y = 2$ by elimination.
Adding the equations: $3x = 9$, so $x = 3$; then $y = 7 - 2(3) = 1$. Solution: $(3, 1)$.
- 13 Find the point(s) of intersection of $y = x^2$ and $y = 2x + 3$.
 $x^2 = 2x + 3$: $x^2 - 2x - 3 = 0$, $(x - 3)(x + 1) = 0$: $x = 3, -1$. Points: $(3, 9)$ and $(-1, 1)$.
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Graphing functions calculus will build on

- 14 The graph of $f(x) = x^3 - 3x$ is shown. Use it to estimate the x -intercepts and the local max/min locations — features calculus will let you find exactly.
From the graph, x -intercepts near $x = 0, \pm 1.7$; local max near $x = -1$, local min near $x = 1$ — these visual estimates match what derivative techniques will later confirm exactly.
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Average rate of change (a preview of the derivative)

- 15 Find the average rate of change of $f(x) = x^2$ over $[1, 4]$, and interpret this as the slope of a secant line.
 $\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = 5$ — this is the slope of the secant line connecting $(1, 1)$ and $(4, 16)$.
- 16 For $f(x) = x^2$, find the average rate of change over $[1, 1 + h]$ in terms of h , and describe what happens as $h \rightarrow 0$ (a preview of the derivative).
 $\frac{f(1+h) - f(1)}{h} = \frac{(1+h)^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h$. As $h \rightarrow 0$, this approaches 2 — the instantaneous rate of change (derivative) of f at $x = 1$.
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Radicals and rational exponents

17 Simplify each expression, writing your answer with positive exponents:

a $x^{1/2} \cdot x^{1/3} = x^{5/6}$

b $(x^{2/3})^3 = x^2$

c $\frac{x^{5/2}}{x^{1/2}} = x^2$

18 Rewrite $\sqrt[3]{x^2}$ using rational exponent notation, and evaluate it at $x = 8$.
 $\sqrt[3]{x^2} = x^{2/3}$. At $x = 8$: $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$.

Interval notation and set operations review

19 Write the solution set of $x^2 - 4 > 0$ in interval notation.

$x^2 > 4 \Rightarrow x < -2$ or $x > 2$. Interval: $(-\infty, -2) \cup (2, \infty)$.